

Pandit Deendayal Energy University

Mid Semester Examination: March 2025
B. Tech. (Computer Science and Engineering)

Semester - IV

Roll No. _____

Date: 04/03/2025
Time: 4 PM – 5 PM
Max. Marks: 25

Course Name: Design and Analysis of Algorithms
Course Code: 20CP209T

Instructions:

1. Do not write anything other than your roll number on question paper.
2. Read question paper properly and assume suitable data wherever essential by mentioning it clearly.
3. Answers written using PENCIL will NOT be considered for evaluation.

Que. No.	Description	Marks	CO
Q1	Select the correct option (1 Mark) and give proper justification (1 Marks) for each of the following.		
Q1 (A)	Consider the following fragment of C-program <pre>for(int i = n, j = 0; i > 0; i /= 2, j += i);</pre> Let A denotes the value of variable j after the termination of for loop, then A = i. $\theta(\log n)$ ii. $\theta(n)$ iii. $\theta(\sqrt{n})$ iv. $\theta(n \log n)$	2	CO-1
Q1 (B)	Which one of the following best represents the approximate value returned by the below given "Hard(n)" function? <pre>int Hard(int n) { int a, b, c, d, e = 0; for (a = 1; a < n; ++a) { d = 0; for (b = n; b > 1; b = b/2) ++d; for (c = 1; c < d; c = c + 2) ++e; } return e; }</pre> i. n^2 ii. $n \log n$ iii. $n \log \log n$ iv. $n(\log n)^2$	2	CO-1
Q1 (C)	The fewest number of comparisons required to check whether an integer appears more than $n/2$ times in a sorted array of n integers in the worst case is: i. $\theta(\log n)$ ii. $\theta(1)$ iii. $\theta(n)$ iv. $\theta(n \log n)$	2	CO-1

Q1 (D)	The solution to the recurrence relation $T(n) = 2T(n/4) + n^{0.51}$ using the Masters Theorem is: i. $\theta(n^{0.50})$ ii. $\theta(n^{0.51})$ iii. $\theta(n^2)$ iv. Master Theorem does not apply	2	CO-2
Q1 (E)	The number of distinct minimum-weight spanning trees of the undirected graph given by the following weight matrix is - $W = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 1 & 2 \\ 1 & 0 & 3 & 0 & 0 & 0 & 2 \\ 0 & 3 & 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 2 & 0 \end{bmatrix}$ i. 1 ii. 6 iii. 9 iv. 15	2	CO-3
Q2	Suppose we perform a sequence of n operations on a data structure in which the i^{th} operation costs i if i is an exact power of 2, and 1 otherwise. Use aggregate analysis to determine the amortized cost per operation.	5	CO-1
Q3	Apply the recurrence tree method to the following equation to obtain the guess: $T(n) = T(n/2) + T(2n/5) + 7n$ Prove that your guess is correct using mathematical induction. OR Obtain the guess to the following recurrence equation using any method: $T(n) = \begin{cases} 2T(\sqrt{n}) + 1, & n > 2 \\ 2, & n \leq 2 \end{cases}$ Prove that your guess is correct using mathematical induction.	5	CO-2
Q4	Consider the following message that needs to be encoded into the binary code. t q s p r t t r q s r t p q t s q s t r t r p s q t The binary code should satisfy the following properties: a. The code assigned to each letter must be unique, ensuring that no letter's code is a prefix of another letter's code. b. If two letters have same frequency, the letter that appears earlier in the dictionary should be assigned a code that is no longer than the one assigned to the other letter. Find the binary code that satisfies the above properties for all the letters in the message using a greedy approach. Also, find the percentage savings using this code when compared with the ASCII code.	5	CO-3

===== End of the Question Paper =====

===== Best of Luck =====